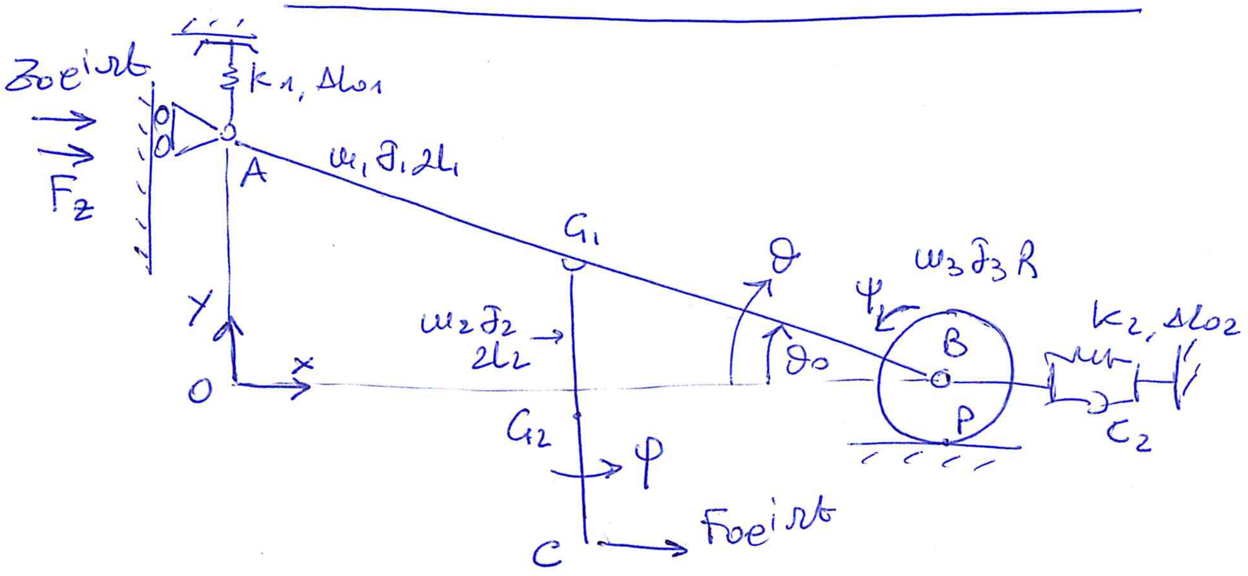
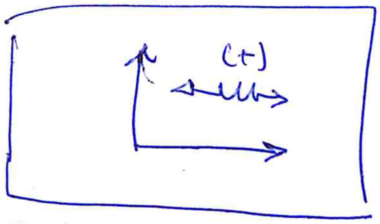
WRITTEN TEST 24TH Nov 2016 - A



- 1 cart, 2 hinges, 1 rolling without sliding condition → 7 d.o.c.
- 3 rigid bodies → 9 d.o.f.

$\Rightarrow$ 2 d.o.f. $\qquad \underline{x}_F = \begin{Bmatrix} \vartheta \\ \varphi \end{Bmatrix} \quad \underline{x}_c = \{z\} \quad \underline{x} = \begin{Bmatrix} \vartheta \\ \varphi \\ z \end{Bmatrix}$

$$T = \frac{1}{2} m_1 \dot{x}_{G1}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} J_1 \dot{\vartheta}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} m_2 \dot{y}_{G2}^2 + \frac{1}{2} J_2 \dot{\varphi}^2 +$$
$$+ \frac{1}{2} m_3 \dot{x}_B^2 + \frac{1}{2} J_3 \dot{\psi}^2$$



$$D = \frac{1}{2} c_2 \Delta \dot{l}_2^2 \; ; \quad V_{EL} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2$$

$$\delta W = \vec{F_0 e^{i\Omega t}} \cdot \delta \vec{x}_c + \vec{F}_z \cdot \delta \vec{z} = F_0 e^{i\Omega t} \delta x_c + F_z \delta z$$

- non linear kinematic relationships:

$$\begin{cases} x_{G1} = L_1 \cos\vartheta + z \\ y_{G1} = L_1 \sin\vartheta \\ x_{G2} = L_1 \cos\vartheta + L_2 \sin\varphi + z \\ y_{G2} = L_1 \sin\vartheta - L_2 \cos\varphi \\ x_B = 2L_1 \cos\vartheta + z \; ; \; x_{B0} = 2L_1 \cos\vartheta_0 \\ x_C = L_1 \cos\vartheta + 2L_2 \sin\varphi + z \\ \psi = -\dfrac{x_B - x_{B0}}{R} = \dfrac{2L_1}{R}(\cos\vartheta_0 - \cos\vartheta) - \dfrac{z}{R} \end{cases}$$

$$\begin{cases} \Delta l_1 = 2L_1(\sin\vartheta_0 - \sin\vartheta) \\ \Delta l_2 = 2L_1(\cos\vartheta_0 - \cos\vartheta) - z \end{cases}$$

• non linear Jacobian matrices:

$$[\Lambda_m]=\begin{bmatrix} -L_1\,\mathrm{sen}\vartheta & 0 & 1 \\ L_1\cos\vartheta & 0 & 0 \\ 1 & 0 & 0 \\ -L_1\,\mathrm{sen}\vartheta & L_2 & 1 \\ L_1\cos\vartheta & 0 & 0 \\ 0 & 1 & 0 \\ -2L_1\,\mathrm{sen}\vartheta & 0 & 1 \\ \frac{2L_1}{R}\,\mathrm{sen}\vartheta & 0 & -\frac{1}{R} \end{bmatrix}$$

$$[\Lambda_c]=\begin{bmatrix} 2L_1\,\mathrm{sen}\vartheta & 0 & -1 \end{bmatrix}$$

$$[\Lambda_k]=\begin{bmatrix} -2L_1\cos\vartheta & 0 & 0 \\ 2L_1\,\mathrm{sen}\vartheta & 0 & -1 \end{bmatrix}$$

$$[\Lambda_e]=\begin{bmatrix} -L_1\,\mathrm{sen}\vartheta & 2L_2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

• linearized Jacobian matrices:

$$[\Lambda_m]=\begin{bmatrix} -L_1\,\mathrm{sen}\vartheta_0 & 0 & 1 \\ L_1\cos\vartheta_0 & 0 & 0 \\ 1 & 0 & 0 \\ -L_1\,\mathrm{sen}\vartheta_0 & L_2 & 1 \\ L_1\cos\vartheta_0 & 0 & 0 \\ 0 & 1 & 0 \\ -2L_1\,\mathrm{sen}\vartheta_0 & 0 & 1 \\ \frac{2L_1}{R}\,\mathrm{sen}\vartheta_0 & 0 & -\frac{1}{R} \end{bmatrix}$$

$$[\Lambda_c]=\begin{bmatrix} 2L_1\,\mathrm{sen}\vartheta_0 & 0 & -1 \end{bmatrix}$$

$$[\Lambda_k]=\begin{bmatrix} -2L_1\cos\vartheta_0 & 0 & 0 \\ 2L_1\,\mathrm{sen}\vartheta_0 & 0 & -1 \end{bmatrix}$$

$$[\Lambda_e]=\begin{bmatrix} -L_1\,\mathrm{sen}\vartheta_0 & 2L_2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[m_{PH}]=\mathrm{diag}(m_1\; m_1\; J_1\; m_2\; m_2\; J_2\; m_3\; J_3)$$

$$[c_{PH}]=\mathrm{diag}(c_1)$$

$$[k_{PH}]=\mathrm{diag}(k_1\; k_2)$$

$$\underline{F}=\{F_0 e^{i\Omega t};\ F_z\}^T$$

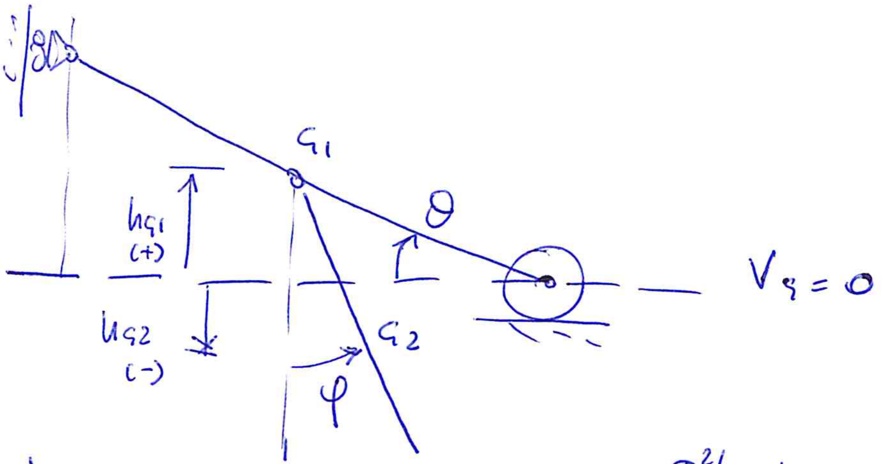
• 2^{nd} order matrices :

$$\frac{\partial^2 \Delta l_1}{\partial \theta^2} = 2L_1 \operatorname{sen}\theta \rightarrow \left.\frac{\partial^2 \Delta l_1}{\partial \theta^2}\right|_{\underline{x}_0} = 2L_1 \operatorname{sen}\theta_0$$

$$\frac{\partial^2 \Delta l_2}{\partial \theta^2} = 2L_1 \cos\theta \rightarrow \left.\frac{\partial^2 \Delta l_2}{\partial \theta^2}\right|_{x_0} = 2L_1 \cos\theta_0$$

$$[k_{II,1}] = k_1 \Delta l_{01} \begin{bmatrix} 2L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[k_{II,2}] = k_2 \Delta l_{02} \begin{bmatrix} 2L_1 \cos\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\begin{cases} h_{q1} = L_1 \operatorname{sen}\theta \\ h_{q2} = L_1 \operatorname{sen}\theta - L_2 \cos\varphi \end{cases} \rightarrow \begin{cases} \left.\frac{\partial^2 h_{q1}}{\partial \theta^2}\right|_{x_0} = -L_1 \operatorname{sen}\theta_0 \\ \left.\frac{\partial^2 h_{q2}}{\partial \theta^2}\right|_{x_0} = -L_1 \operatorname{sen}\theta_0 \, ; \, \left.\frac{\partial^2 h_{q2}}{\partial \varphi^2}\right|_{x_0} = L_2 \end{cases}$$

$$[k_{q1}] = m_1 g \begin{bmatrix} -L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} ; [k_{q2}] = m_2 g \begin{bmatrix} -L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- structural matrices:

$$[M] = [\Lambda_m]^T [m_{PH}] [\Lambda_m]$$
$$[C] = [\Lambda_c]^T [C_{PH}] [\Lambda_c]$$
$$[k] = [\Lambda_k]^T [k_{PH}] [\Lambda_k] + [k_{\Pi,1}] + [k_{\Pi,2}] + [k_{s1}] + [k_{s2}]$$

- matrices partition:

$$\begin{cases} M_{FF} = M(1:2,\ 1:2) \\ M_{FC} = M(1:2,\ 3) \end{cases}$$ the same for $[C]$ and $[k]$

$$\underline{Q} = \begin{Bmatrix} \underline{Q}_F \\ \hline \underline{Q}_c \end{Bmatrix} = (\Lambda_Q)^T \underline{F} \rightarrow \underline{Q}_F = \begin{Bmatrix} -L_1 \operatorname{sen}\vartheta_0 \\ 2L_2 \end{Bmatrix} F_0 e^{i\Omega t}$$

$$\underline{Q}_c = \{ F_0 e^{i\Omega t} + F_2 \}$$

- question nr. 4:

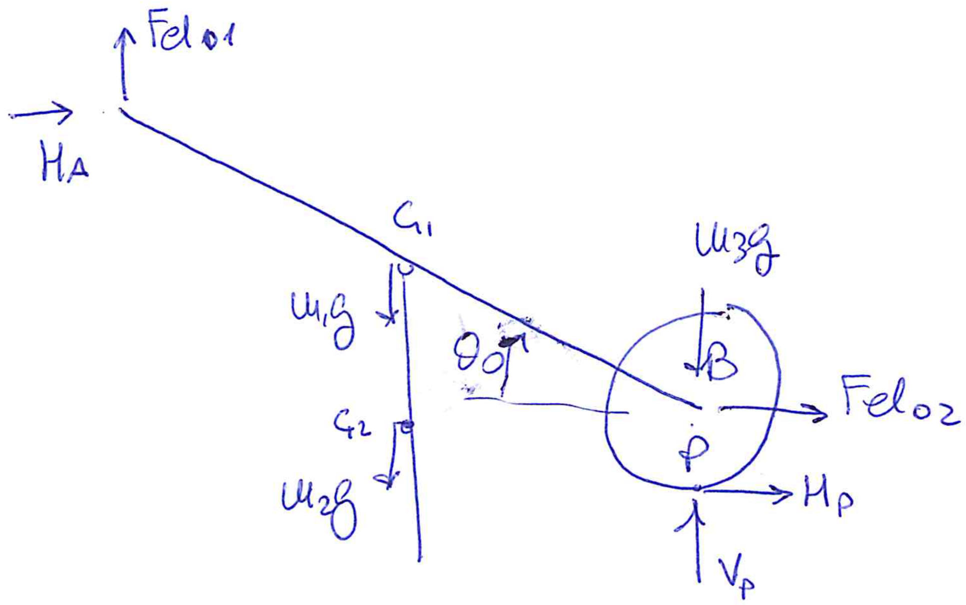
$$\begin{cases} F_0 \neq 0 \\ z_0 = 0 \end{cases} \rightarrow \begin{array}{l} \underline{Q}_F = \underline{Q}_{F0} e^{i\Omega t} \ ; \ \underline{Q}_{F0} = \begin{Bmatrix} -L_1 \operatorname{sen}\vartheta_0 \\ 2L_2 \end{Bmatrix} F_0 \\ \underline{x}_c = 0 \end{array}$$

- question nr. 5:

$$\begin{cases} F_0 = 0 \\ z_0 \neq 0 \end{cases} \rightarrow \begin{array}{l} \underline{Q}_F = 0 \\ \underline{Q}_{FC} = -\{ -\Omega^2 [M_{FC}] + i\Omega [C_{FC}] + [k_{FC}] \} \underline{x}_{c0} \end{array}$$

$$\underline{x}_c = \underline{x}_{c0} e^{i\Omega t} \ ; \ \underline{x}_{c0} = \{ z_0 \}$$

· question nr. 6



$$\sum M_B^{(DISK)} = 0 \quad : \quad H_P = 0$$

$$\sum F_x^{(ALL)} = 0 \quad : \quad H_A + F_{elo2} = 0 \rightarrow H_A = -F_{elo2}$$

$$\sum M_B^{(ALL)} = 0 \quad : \quad F_{elo1}\, 2\cancel{L_1} \cos\theta_0 - (m_1 + m_2) g \cancel{L_1} \cos\theta_0 +$$

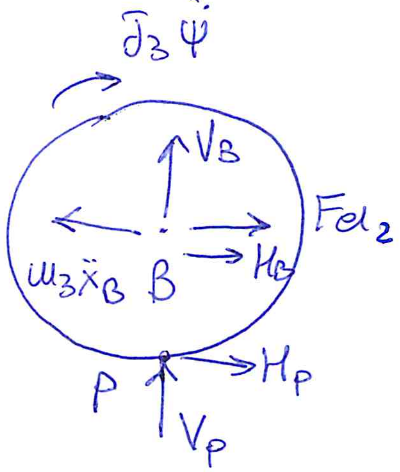
$$+ H_A\, 2\cancel{L_1} \operatorname{sen}\theta_0 = 0 \rightarrow$$

$$\rightarrow F_{elo1} = \frac{(m_1 + m_2) g}{2} + F_{elo2} \operatorname{Tg}\theta_0$$

$$F_{elo2} = k_2 \Delta l_{o2}$$

$$\Delta l_{o1} = \frac{F_{elo1}}{k_1}$$

• Question nr. 7

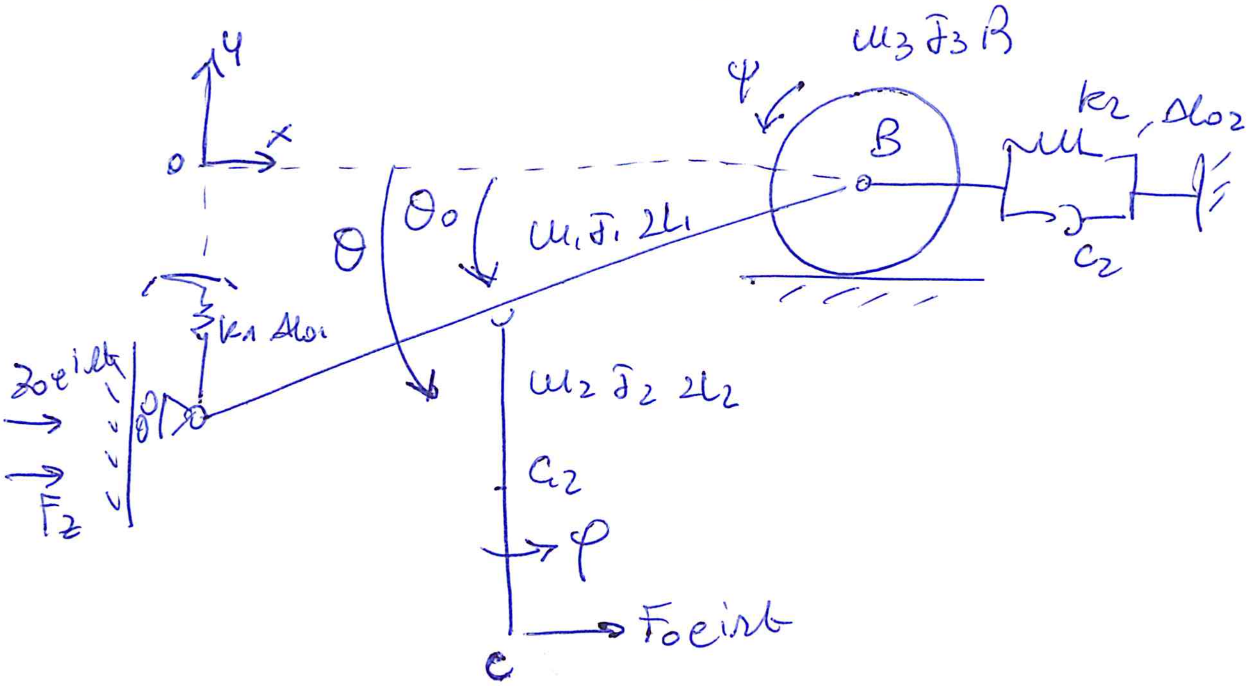


$$\sum M_P^{DISK} = 0 : \quad Fel_2 R + H_B R - m_3\ddot{x}_B R + J_3\ddot{\psi} = 0 \rightarrow$$

$$\rightarrow \quad H_B = m_3\ddot{x}_B - \frac{J_3\ddot{\varphi}}{R} - Fel_2$$

$$Fel_2 = k_2\Delta l_2 + c_2\Delta\dot{l}_2$$

VERSION B



Diferences with respect of the solution of version A:

$$x_{G1} = L_1 \cos\theta + z$$

$$y_{G1} = -L_1 \,\text{sen}\,\theta$$

$$x_{G2} = L_1 \cos\theta + L_2 \,\text{sen}\,\varphi + z$$

$$y_{G2} = -L_1 \,\text{sen}\,\theta - L_2 \cos\varphi$$

$$x_B = 2L_1 \cos\theta + z \quad ; \quad x_{B0} = 2L_1 \cos\theta_0$$

$$\psi = -\frac{x_B - x_{B0}}{R} = \frac{2L_1}{R}(\cos\theta_0 - \cos\theta) - \frac{z}{R}$$

$$\Delta l_1 = 2L_1 (\text{sen}\,\theta - \text{sen}\,\theta_0)$$

$$\Delta l_2 = 2L_1 (\cos\theta_0 - \cos\theta) - z$$

· linearized Jacobian matrices:

$$[\Lambda_m] = \begin{bmatrix} -L_1 \operatorname{sen}\theta_0 & 0 & 1 \\ -L_1 \cos\theta_0 & 0 & 0 \\ 1 & 0 & 0 \\ -L_1 \operatorname{sen}\theta_0 & L_2 & 1 \\ -L_1 \cos\theta_0 & 0 & 0 \\ 0 & 1 & 0 \\ -2L_1 \operatorname{sen}\theta_0 & 0 & 1 \\ +\frac{2L_1}{R} \operatorname{sen}\theta_0 & 0 & -\frac{1}{R} \end{bmatrix}$$

$$[\Lambda_c] = \begin{bmatrix} 2L_1 \operatorname{sen}\theta_0 & 0 & -1 \end{bmatrix}$$

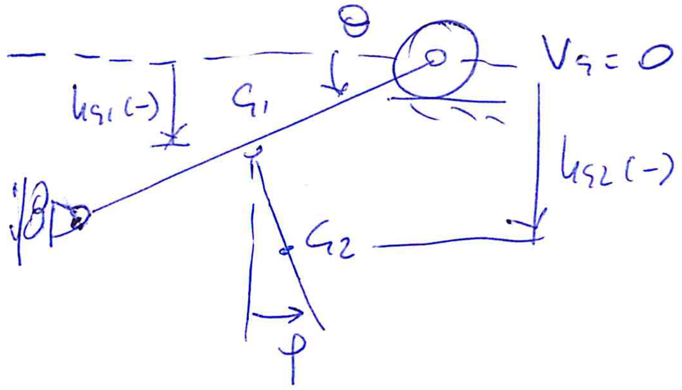
$$[\Lambda_k] = \begin{bmatrix} 2L_1 \cos\theta_0 & 0 & 0 \\ 2L_1 \operatorname{sen}\theta_0 & 0 & -1 \end{bmatrix}$$

$$[\Lambda_Q] = \begin{bmatrix} -L_1 \operatorname{sen}\theta_0 & 2L_2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

· 2nd order stiffness matrices:

$$\left.\frac{\partial^2 \Delta l_1}{\partial \theta^2}\right|_{x_0} = -2L_1 \operatorname{sen}\theta_0 \rightarrow [k_{\Pi,1}] = k_1 \Delta l_{01} \begin{bmatrix} -2L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

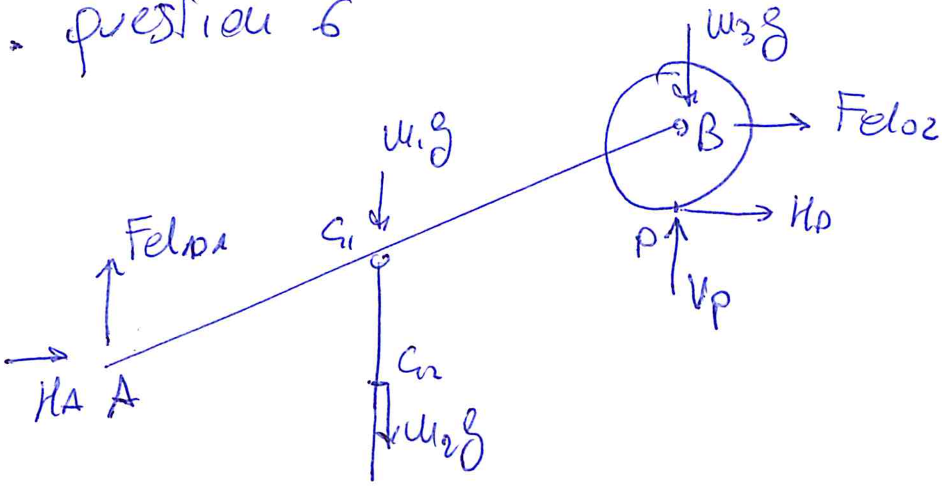
$$\left.\frac{\partial^2 \Delta l_2}{\partial \theta^2}\right|_{x_0} = 2L_1 \cos\theta_0 \rightarrow [k_{\Pi,2}] = k_2 \Delta l_{02} \begin{bmatrix} 2L_1 \cos\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\begin{cases} h_{q1} = -L_1 \operatorname{sen}\theta \\ h_{q2} = -L_1 \operatorname{sen}\theta - L_2 \cos\varphi \end{cases}$$

$$[k_{q1}] = m_1 g \begin{bmatrix} L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \quad [k_{q2}] = m_2 g \begin{bmatrix} L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & L_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

• question 6



same equations of version A ...

... $H_P = 0$

$H_A = -F_{elo2}$

$F_{elo1}\, 2L_1 \cos\vartheta_0 - (m_1 + m_2) g\, L_1 \cos\vartheta_0 - H_A\, 2L_1 \operatorname{sen}\vartheta_0 = 0$

$\rightarrow F_{elo1} = \frac{m_1 + m_2}{2} g - F_{elo2} \operatorname{Tg} \vartheta_0$